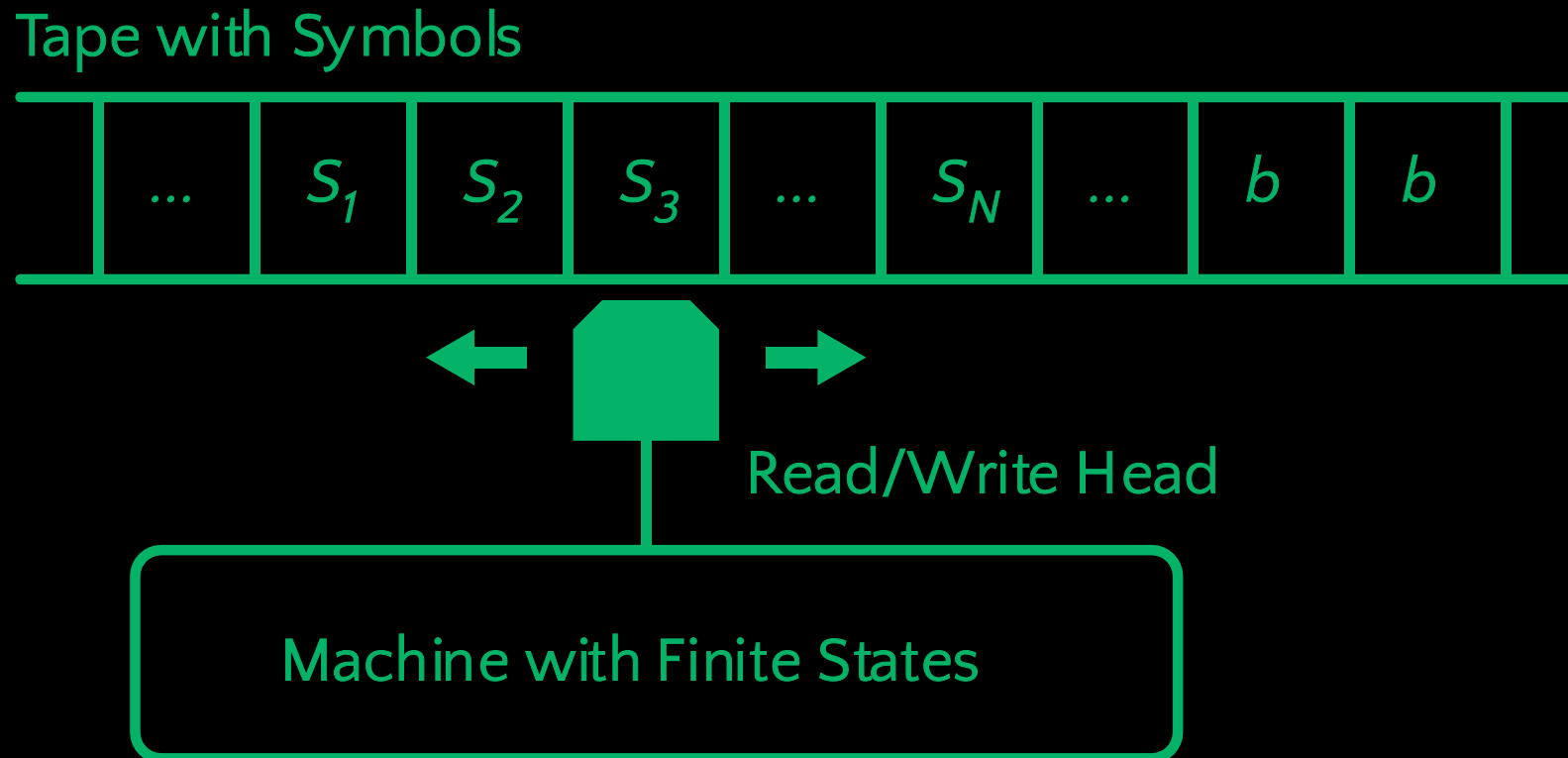


Computation and Abstraction

Dr. Changlai Du

Is 35500001 divisible by 113?

The Turing Machine



Transition Function

$$\delta(q, a) = (q', b, D)$$

$q_0 3 5 5 0 0 0 0 1 \rightarrow 0 q_3 5 5 0 0 0 0 1$
 $\rightarrow 0 0 q_{35} 5 0 0 0 0 1$
 $\rightarrow 0 0 3 q_{16} 0 0 0 0 1$
 $\rightarrow 0 0 3 1 q_{47} 0 0 0 1$
 $\rightarrow 0 0 3 1 4 q_{18} 0 0 1$
 $\rightarrow 0 0 3 1 4 1 q_{67} 0 1$
 $\rightarrow 0 0 3 1 4 1 5 q_{105} 1$
 $\rightarrow 0 0 3 1 4 1 5 9 q_{34-}$

What is the greatest common divisor of 96 and 404?

The Universal Turing Machine

$$U(\langle M \rangle, x) = M(x)$$

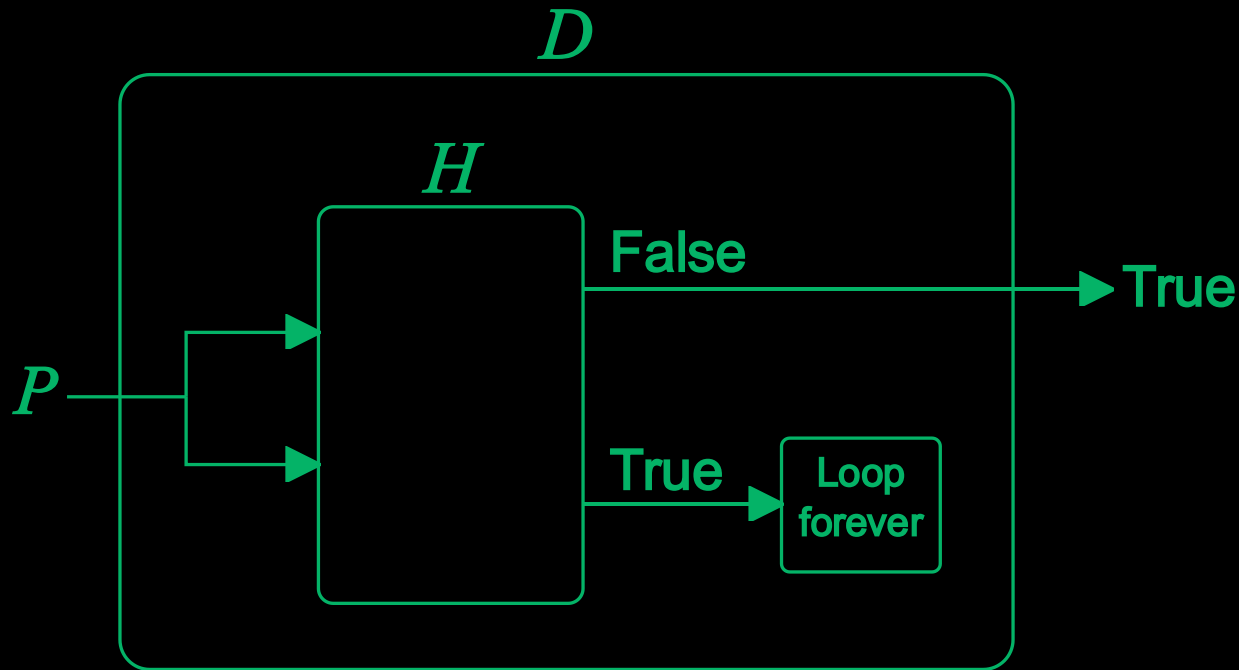
The Church–Turing Thesis

Every function that is effectively computable by an algorithm can be computed by a Turing machine.

Can every clearly defined problem be solved by such a machine?

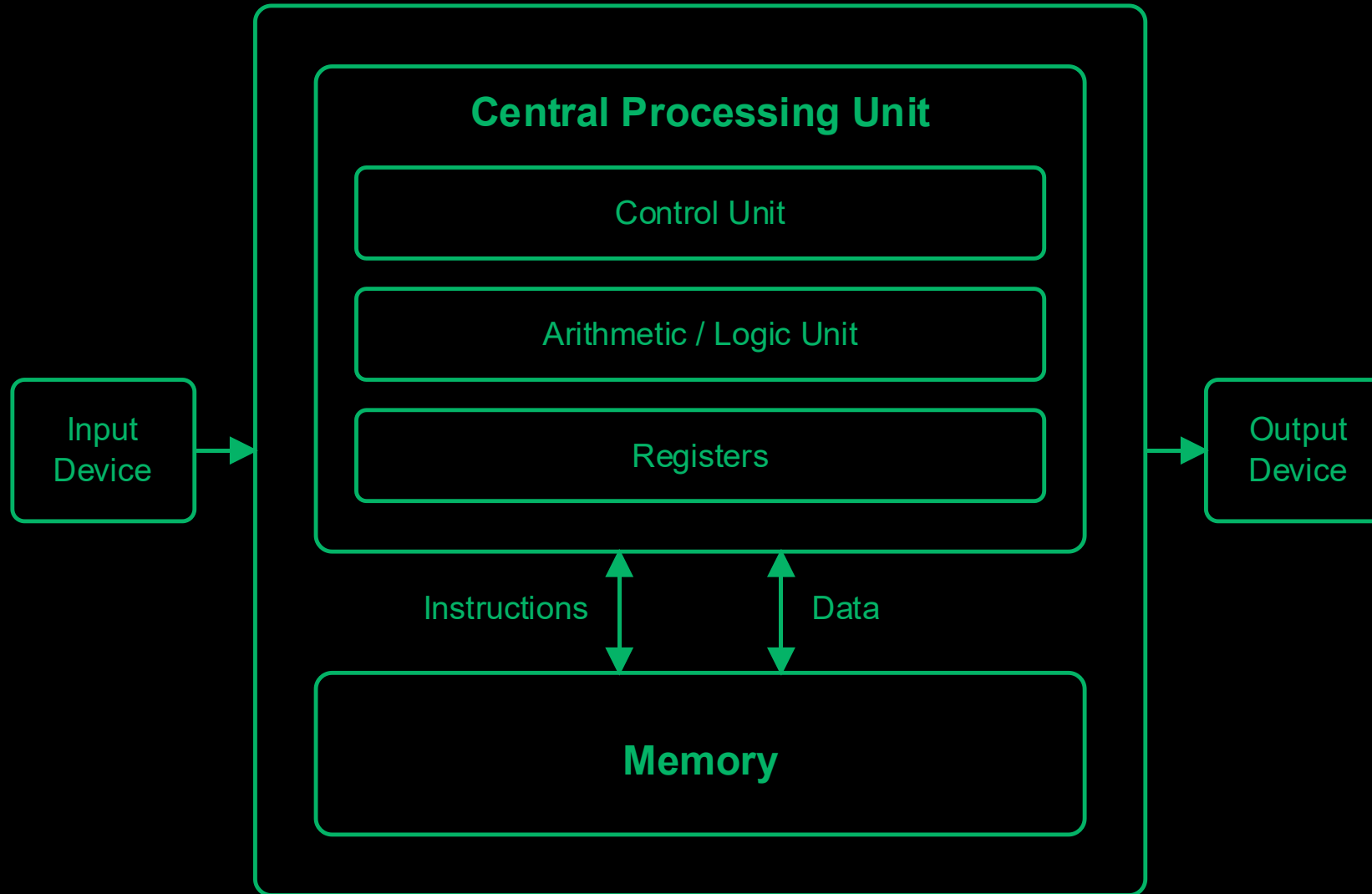
The Turing Halting Problem

Can a universal procedure determine whether any given program will eventually stop when run on a given input?



```
def D():  
    if H(D):  
        loop_forever()
```

Von Neumann Architecture



How many names did you check to spot yourself on the attendance sheet?

What is the time and space complexity computing the Fibonacci number $f(n)$?

“We admire Wiles’s proof of Fermat’s Last Theorem, the scientific theories of Newton, Einstein, Darwin, Watson and Crick, and the designs of the Golden Gate Bridge and the pyramids precisely because they seem to require a leap that cannot be made by everyone, let alone by a simple mechanical device.”

— Avi Wigderson

and more...

Concepts Checklist

instructions	Euclidean Algorithm	complexity
algorithm	simulation	big-O
input	Stored-Program Computer	recursion
output	functions	dynamic programming
decision problems	thesis	time complexity
search problems	Church-Turing Thesis	space complexity
model	Turing complete	P vs NP
mechanical	computable numbers	complexity zoo
intelligence	Halting Problem	abstraction
Turing Test	decidability	interface
Turing machine	Von Neumann Architecture	implementation
computability	program counter	

Homework 1

Watch the Week 0 video of CS50

<https://cs50.harvard.edu/x/weeks/0/>

Homework 2
Watch the videos
Map of Computer Science
and
From Nand to Tetris in 12 steps